ZHU WANG

zhu.wang@kellogg.northwestern.edu Personal Website Github Google Scholar

RESEARCH INTERESTS

My research advances multimodal AI for reasoning, adaptation, and discovery. I focus on structured knowledge and visual reasoning, efficient and adaptive AI, and human-AI creativity and technological innovation.

EDUCATION

University of Illinois at Chicago, Chicago, IL 2025
Ph.D., Computer Science
Advisors: Sathya N. Ravi, Isabel Cruz (*in memoriam*)
Thesis: Semantic-driven Multimodal Learning: Gradient-based Methods and Applications

University of Pittsburgh, Pittsburgh, PA 2015
M.S., Information Science

Northeastern University, China 2012
B.Mgt., Information Management and System

RESEARCH EXPERIENCE

Research Assistant Professor, NICO, Northwestern University 07/2026 - present
Postdoctoral Fellow, NICO, Northwestern University 02/2025 - 06/2026
Supervisor: Prof. Brian Uzzi
Visualizing Innovation: Multimodal Analysis of Scientific Evolution

Research Assistant, AIOpt Lab, University of Illinois at Chicago 01/2022 - 12/2024
Multimodal Representation Learning and Applications

Research & Data Science Intern, CCC Intelligence Solutions, Chicago, IL 05/2022 - 08/2022

Research Assistant, ADVIS Lab, University of Illinois at Chicago 08/2016 - 10/2021
Knowledge Graph Embedding; Deep Ontology Matching; Machine Learning for Public Health

Research Assistant, PICSO Lab, University of Pittsburgh 12/2014 - 04/2016
Scholarly Knowledge Link Prediction

Undergraduate Research Assistant, Northeastern University, China 09/2011 - 06/2012
Human-machine Decision-making System

PUBLICATIONS AND MANUSCRIPTS

Bold *indicates my name. * indicates the student(s) I mentor.*

Peer-Reviewed Journal/Conference Papers:

[J1] Targeted Unlearning Using Perturbed Sign Gradient Methods With Applications on Medical Images
*Transactions on Machine Learning Research (***TMLR), 2025.**
George Nahass*, **Zhu Wang**, Homa Rashidisabet, Won Hwa Kim, Sasha Hubschman, Jeffrey C Peterson, Ghasem Yazdanpanah, Chad A Purnell, Pete Setabutr, Ann Q Tran, Darvin Yi, Sathya N. Ravi

[c2] Inventive Problem Solving with LLMs: A Benchmark for TRIZ Reasoning
*The 64rd Annual Meeting of the Association for Computational Linguistics (***Findings of ACL 2026).**
Zhu Wang, Brian Uzzi

[c3] DesignCLIP: Multimodal Learning with CLIP for Design Patent Understanding
*Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (***Findings of**

EMNLP 2025).
Zhu Wang, Homaira Huda Shomee*, Sathya N. Ravi, Sourav Medya
Model release: **10K+ downloads on Hugging Face**

[c4] **From Heart to Words: Generating Empathetic Responses via Integrated Figurative Language and Semantic Context Signals**
*The 63rd Annual Meeting of the Association for Computational Linguistics (***Findings of ACL 2025).**
Gyeongeun Lee*, **Zhu Wang**, Sathya N. Ravi, Natalie Parde

[c5] **A Survey on Patent Analysis: From NLP to Multimodal AI**
*The 63rd Annual Meeting of the Association for Computational Linguistics (***ACL 2025).**
Homaira Huda Shomee*, **Zhu Wang**, Sathya N. Ravi, Sourav Medya

[c6] **Optimizing Neural Network Training and Quantization with Rooted Logistic Objectives**
*The 28th International Conference on Artificial Intelligence and Statistics (***AISTATS 2025).**
Zhu Wang, Praveen Raj Veluswami*, Harsh Mishra*, Sathya N. Ravi

[c7] **IMPACT: A Large-scale Integrated Multimodal Patent Analysis and Creation Dataset for Design Patents**
*Thirty-eighth Conference on Neural Information Processing Systems (***NeurIPS 2024).**
Homaira Huda Shomee*, **Zhu Wang**, Sathya N. Ravi, Sourav Medya

[c8] **Implicit Differentiable Outlier Detection Enable Robust Deep Multimodal Analysis**
*Thirty-seventh Conference on Neural Information Processing Systems (***NeurIPS 2023).**
Zhu Wang, Sourav Medya, Sathya N. Ravi
TL;DR: An end-to-end VLM framework incorporating explicit knowledge graphs and OOD-detection.

[c9] **Contextualized Structural Self-supervised Learning for Ontology Matching**
*The 18th International Workshop on Ontology Matching (***OM@ISWC 2023).**
Zhu Wang

Peer-Reviewed Workshop Papers:
[w10] **EmpatheticFIG at WASSA 2024 Empathy and Personality Shared Task: Predicting Empathy and Emotion in Conversations with Figurative Language**
*Workshop on Computational Approaches to Subjectivity, Sentiment, & Social Media Analysis at (***ACL 2024).**
Gyeongeun Lee*, **Zhu Wang**, Sathya N. Ravi, Natalie Parde

[w11] **Analysis of the impact of covid-19 on education based on geotagged twitter**
*Workshop on Modeling and Understanding the Spread of COVID-19 (***SIGSPATIAL 2020).**
Zhu Wang, Isabel Cruz

[w12] **Predictive analytics using text classification for restaurant inspections**
*Workshop on Smart Cities and Urban Analytics (***SIGSPATIAL 2017).**
Zhu Wang, Booma Sowkarthiga Balasubramani, Isabel Cruz

[w13] **Research on scenarios management for testing decision-making behavior in decision simulation environment**
*International Conference on Systems and Informatics (***ICSAI 2012).**
Zhu Wang, Zhuchao Yu, Yang Jiang, Xueying Hong

[s14,15] **AgreementMakerDeep Results for OAEI 2021, 2022**
*The 16th/17th International Workshop on Ontology Matching (***OM@ISWC 2021, 2022).**
Zhu Wang, Isabel Cruz

MANUSCRIPTS UNDER REVIEW:
[u1] **Text-Conditioned Diffusion Cuts for Training-Free Open-Vocabulary Segmentation**
Zhu Wang, Harsh Mishra*, Sathya N. Ravi

[u2] Credible Synthesis or Off-Rail Novelty? Evaluating LLM-Generated Interdisciplinary Research Ideas
Huilian Sophie Qiu, **Zhu Wang**, Huaxia Zhou, Rui Pan, Hongbo Fang

[u3] Every Head Counts: Mask-Alignment Calibration for Semantic Recovery Under Domain Shift
Nicolò Vacis*, **Zhu Wang**, Yuyao Yang, Giovanni Vaccarino*, Sathya N. Ravi

MANUSCRIPTS IN PREPARATION:

[p1] How Technology Booms Take Shape: The Structure and Dynamics of Technological Innovation
Zhu Wang, Brian Uzzi

[p2] Connecting Geometry, Topology, and Logic Enable Structural Visual Reasoning
Varun Hasso Viges*, **Zhu Wang**, Brian Uzzi

[p3] From Sketches to 3D: Generation through Multimodal Guidance
Rasleen Dhaliwal*, Daksh Soni*, Manav Malik*, **Zhu Wang**, Sourav Medya

[p4] FinAffect: Evaluating Emotional Influences on LLM Financial Decisions
Varun Hasso Viges*, Rochana Chaturvedi, **Zhu Wang**, Brian Uzzi, Sourav Medya

TEACHING EXPERIENCE

Delivered Guest Lectures, University of Illinois at Chicago, 08/2019 – 04/2024

- Introduction to Data Science - Topics on Data Preprocessing and Exploratory Data Analysis
- Introduction to Machine Learning - Topics on Decision Trees and Regression Models
- Data and Web Semantics - Topics on Ontology Engineering
- Deep Learning for Computer Vision - Topics on Vision-Language Models

Teaching Assistant, University of Illinois at Chicago, 08/2016 – 12/2024
Course list: Program Design I, Program Design II, Introduction to Data Science, Introduction to Machine Learning, Database Systems, Data and Web Semantics
Teaching Assistant, University of Pittsburgh, **Course:** Data Mining 2015

MENTORING EXPERIENCE

PhD Students:

- Gyeongeun Lee (UIC CS) - Empathetic detection and response generation
- Homaira Huda Shomee (UIC CS) - Multimodal patent analysis
- George Nahass (UIC Medical School) - Medical imaging

CS Master students: UIC: Praveen Raj Veluswami, Harsh Mishra, Raj Mehta, Giovanni Vaccarino, Nicolo Vacis (co-advise on master thesis), Sai Chittala; **Northwestern:** Medini Chopra
Undergraduate Students: UIC CS: Yiming Wang, Angela Xu, Jenny Vuong, Brian Wilk, Rasleen Dhaliwal, Daksh Soni, Manav Malik; **Northwestern ECE:** Varun Hasso Viges

INVITED TALKS & POSTER

Breakthroughs Don't Grow Alone: How Technology Booms Take Shape
2027 AAAS Annual Meeting, to be held February 18 - 20, 2027 in Chicago, IL
Rooted Logistic Loss for Efficient Multimodal Learning with Quantization and Pruning
Multimodal Artificial Intelligence Workshop 2024, TTIC, August 2024

Integrating Semantic Insights in Gradient Based Multimodal Learning
The University of Chicago Pritzker School of Medicine, December 2023
Accelerated Neural Network Training with Rooted Losses
2023 Conference on the Mathematical Theory of Deep Neural Networks, November 2023
Multimodal Representation Learning
University of Chicago, October 2023
Differentiable Outlier Detection Enable Robust Deep Multimodal Analysis
Midwest Machine Learning Symposium, May 2023

ACADEMIC SERVICES

Conference Reviewer/Program Committee member: AISTATS 23 - 25, NeurIPS 23 - 25, ICLR 24 - 26, ICML 24 - 25, ECCV 24, MICCAI 24 &25, AAAI 26-27, SDM 25, ICCV 25, WACV 26, CVPR 26, ICML 26 (Gold Reviewer)
Guest Editor/Journal Reviewer: Special issue on Ontology Matching and Machine Learning of the Semantic Web Journal, Journal of Social Computing, Transactions on Machine Learning Research (TMLR)
Conference Volunteer: Midwest Machine Learning Symposium (MMLS), 2023

AWARDS & HONORS

Graduate Student Council Travel Award, University of Illinois at Chicago 2018, 2023
Graduate SIS Scholarship, University of Pittsburgh 2015

REFERENCES

Prof. Sathya N. Ravi
Assistant Professor, Department of Computer Science
University of Illinois Chicago
Email: sathya@uic.edu
Phone: 312-996-2946

Prof. Sourav Medya
Assistant Professor, Department of Computer Science
University of Illinois Chicago
Email: medya@uic.edu
Phone: 312-996-2307

Prof. Brian Uzzi
Richard L. Thomas Professor, Kellogg School of Management
Northwestern University
Email: uzzi@kellogg.northwestern.edu
Phone: 847-491-8072